

JAMF ADCS

Certificate Setup Guide

EAP-TLS Certificate-Based Wireless · Device & Computer Certificates · ADCS + Jamf Pro

AUTHOR

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VERSION

1.0 · March 2018

USE CASE

EAP-TLS Wireless

ITEMS YOU'LL NEED

Certificates (.cer format)

- Root CA Certificate
- Issuing Certificate

Server Access

- Root CA server (to create cert template)

Console Access

- Admin rights to your Jamf Pro console

Network Info

- FQDN of your RADIUS server(s)

SECTIONS COVERED

- Step 1: General
- Request Handling
- Key Attestation
- Extensions: Application Policies
- Extensions: Certificate Template Info
- Extensions: Key Usage
- Server
- Issuance Requirements
- Network: Protocols
- Compatibility
- Cryptography
- Superseded Templates
- Extensions: Basic Constraints
- Extensions: Issuance Policies
- Security
- Subject Name
- Step 2: JAMF Setup
- Network: Trust

Step 1: General

The screenshot shows the 'Jamf Properties' dialog box with the 'General' tab selected. The dialog has a title bar with a question mark and a close button. The main area contains several fields and checkboxes. The 'Template display name' and 'Template name' fields both contain the text 'Jamf'. The 'Validity period' is set to '2 years' and the 'Renewal period' is set to '6 weeks'. There are two checkboxes: 'Publish certificate in Active Directory' (unchecked) and 'Do not automatically reenroll if a duplicate certificate exists in Active Directory' (unchecked). At the bottom, there are four buttons: 'OK', 'Cancel' (highlighted with a blue border), 'Apply', and 'Help'.

Subject Name		Issuance Requirements		
Superseded Templates		Extensions	Security	Server
General	Compatibility	Request Handling	Cryptography	Key Attestation

Template display name:
Jamf

Template name:
Jamf

Validity period:
2 years

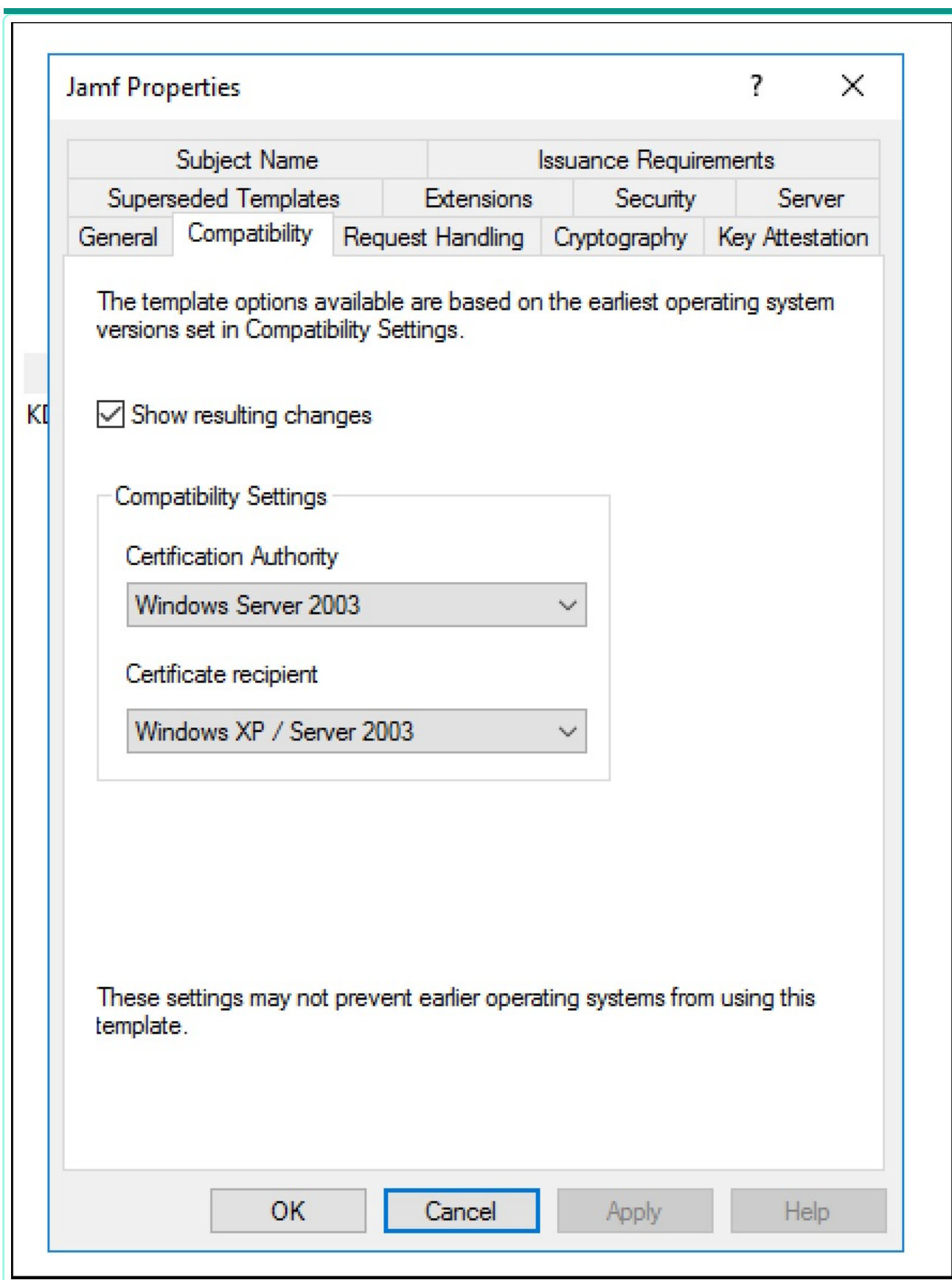
Renewal period:
6 weeks

☐ Publish certificate in Active Directory

☐ Do not automatically reenroll if a duplicate certificate exists in Active Directory

OK Cancel Apply Help

Compatibility



Request Handling

The screenshot shows the 'Jamf Properties' dialog box with the 'Request Handling' tab selected. The dialog has a title bar with a question mark and a close button. Below the title bar is a tabbed interface with the following tabs: 'Subject Name', 'Issuance Requirements', 'Superseded Templates', 'Extensions', 'Security', 'Server', 'General', 'Compatibility', 'Request Handling' (selected), 'Cryptography', and 'Key Attestation'. The 'Request Handling' tab contains the following controls:

- Purpose:** A dropdown menu set to 'Signature and encryption'.
- ☐ Delete revoked or expired certificates (do not archive)
- ☐ Include symmetric algorithms allowed by the subject
- ☐ Archive subject's encryption private key
- ☐ Authorize additional service accounts to access the private key (*)
 Key Permissions...
- ☒ Allow private key to be exported
- ☐ Renew with the same key (*)
- ☐ For automatic renewal of smart card certificates, use the existing key if a new key cannot be created (*)
- Do the following when the subject is enrolled and when the private key associated with this certificate is used:**
 - ☒ Enroll subject without requiring any user input
 - ☐ Prompt the user during enrollment
 - ☐ Prompt the user during enrollment and require user input when the private key is used
- * Control is disabled due to [compatibility settings](#).

At the bottom of the dialog are four buttons: 'OK', 'Cancel' (highlighted with a blue border), 'Apply', and 'Help'.

Cryptography

The screenshot shows the 'Jamf Properties' dialog box with the 'Cryptography' tab selected. The dialog has a title bar with a question mark and a close button. The main area contains several sections: 'Subject Name' and 'Issuance Requirements' at the top, followed by 'Superseded Templates', 'Extensions', 'Security', and 'Server'. Below these are tabs for 'General', 'Compatibility', 'Request Handling', 'Cryptography', and 'Key Attestation'. The 'Cryptography' tab is active, showing a 'Provider Category' dropdown set to 'Legacy Cryptographic Service Provider', an 'Algorithm name' dropdown set to 'Determined by CSP', and a 'Minimum key size' text box containing '2048'. Below these is a section titled 'Choose which cryptographic providers can be used for requests' with two radio buttons: 'Requests can use any provider available on the subject's computer' (unselected) and 'Requests must use one of the following providers:' (selected). Under this, a list of providers is shown with checkboxes: 'Microsoft RSA SChannel Cryptographic Provider' (checked), 'Microsoft DH SChannel Cryptographic Provider' (unchecked), 'Microsoft Enhanced Cryptographic Provider v1.0' (unchecked), 'Microsoft Enhanced DSS and Diffie-Hellman Cryptographic Pr' (unchecked), and 'Microsoft Enhanced RSA and AES Cryptographic Provider' (unchecked). To the right of the list are up and down arrow buttons. Below the list is a 'Request hash' dropdown set to 'Determined by CSP' and a checkbox for 'Use alternate signature format' which is unchecked. At the bottom are 'OK', 'Cancel', 'Apply', and 'Help' buttons.

Jamf Properties

Subject Name Issuance Requirements

Superseded Templates Extensions Security Server

General Compatibility Request Handling Cryptography Key Attestation

Provider Category: Legacy Cryptographic Service Provider

Algorithm name: Determined by CSP

Minimum key size: 2048

Choose which cryptographic providers can be used for requests

☐ Requests can use any provider available on the subject's computer

☒ Requests must use one of the following providers:

Providers:

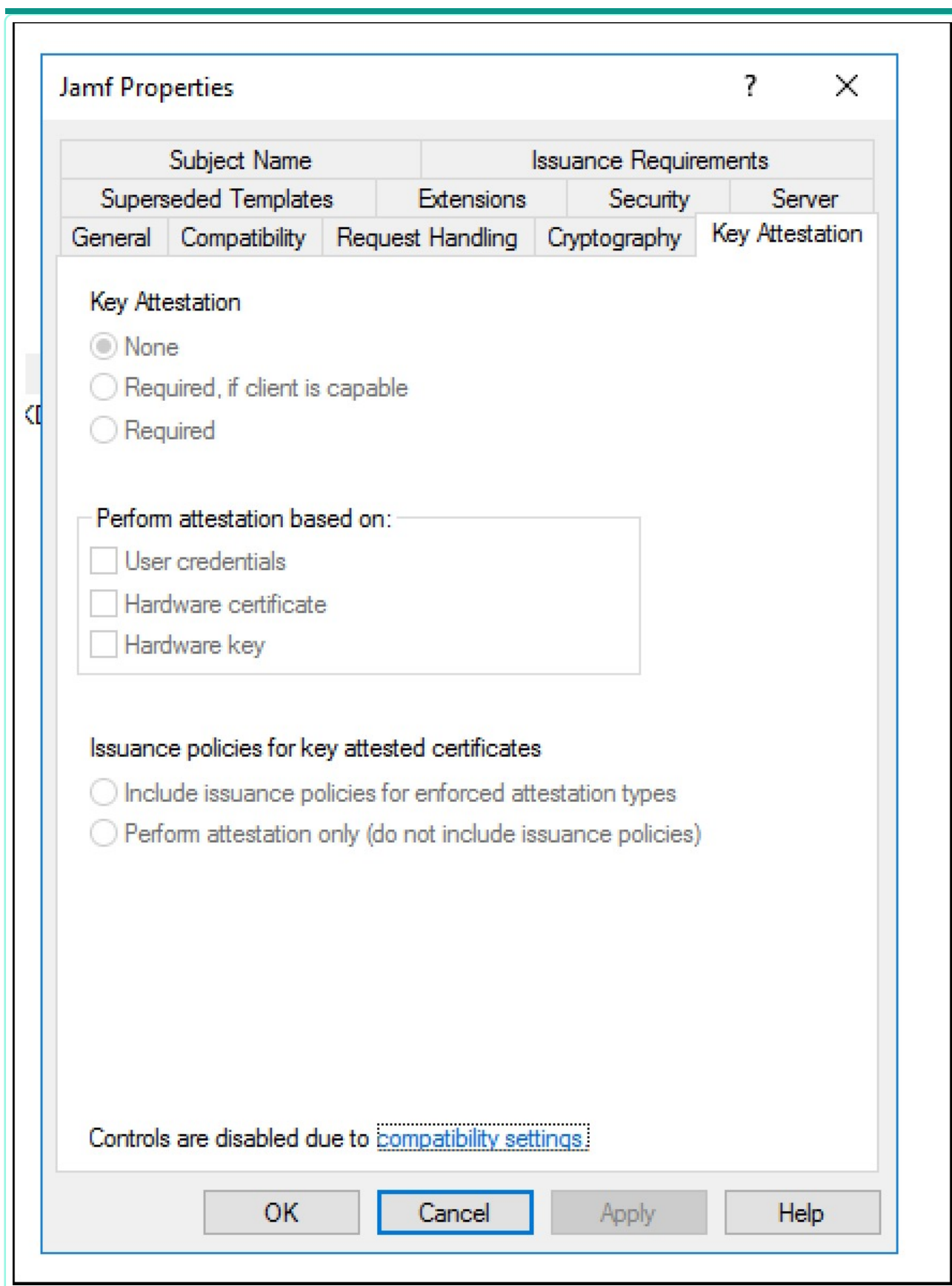
- ☒ Microsoft RSA SChannel Cryptographic Provider
- ☐ Microsoft DH SChannel Cryptographic Provider
- ☐ Microsoft Enhanced Cryptographic Provider v1.0
- ☐ Microsoft Enhanced DSS and Diffie-Hellman Cryptographic Pr
- ☐ Microsoft Enhanced RSA and AES Cryptographic Provider

Request hash: Determined by CSP

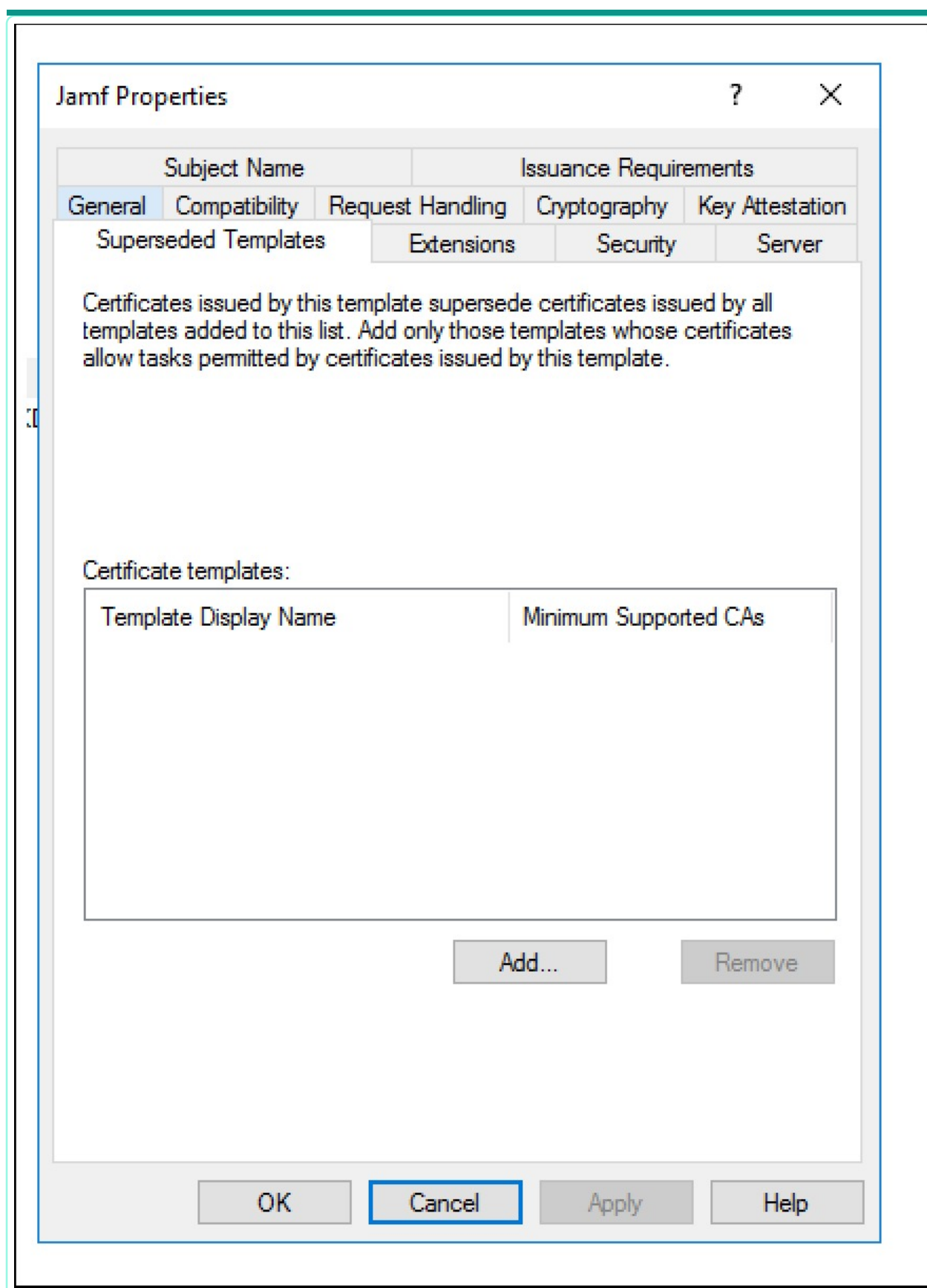
☐ Use alternate signature format

OK Cancel Apply Help

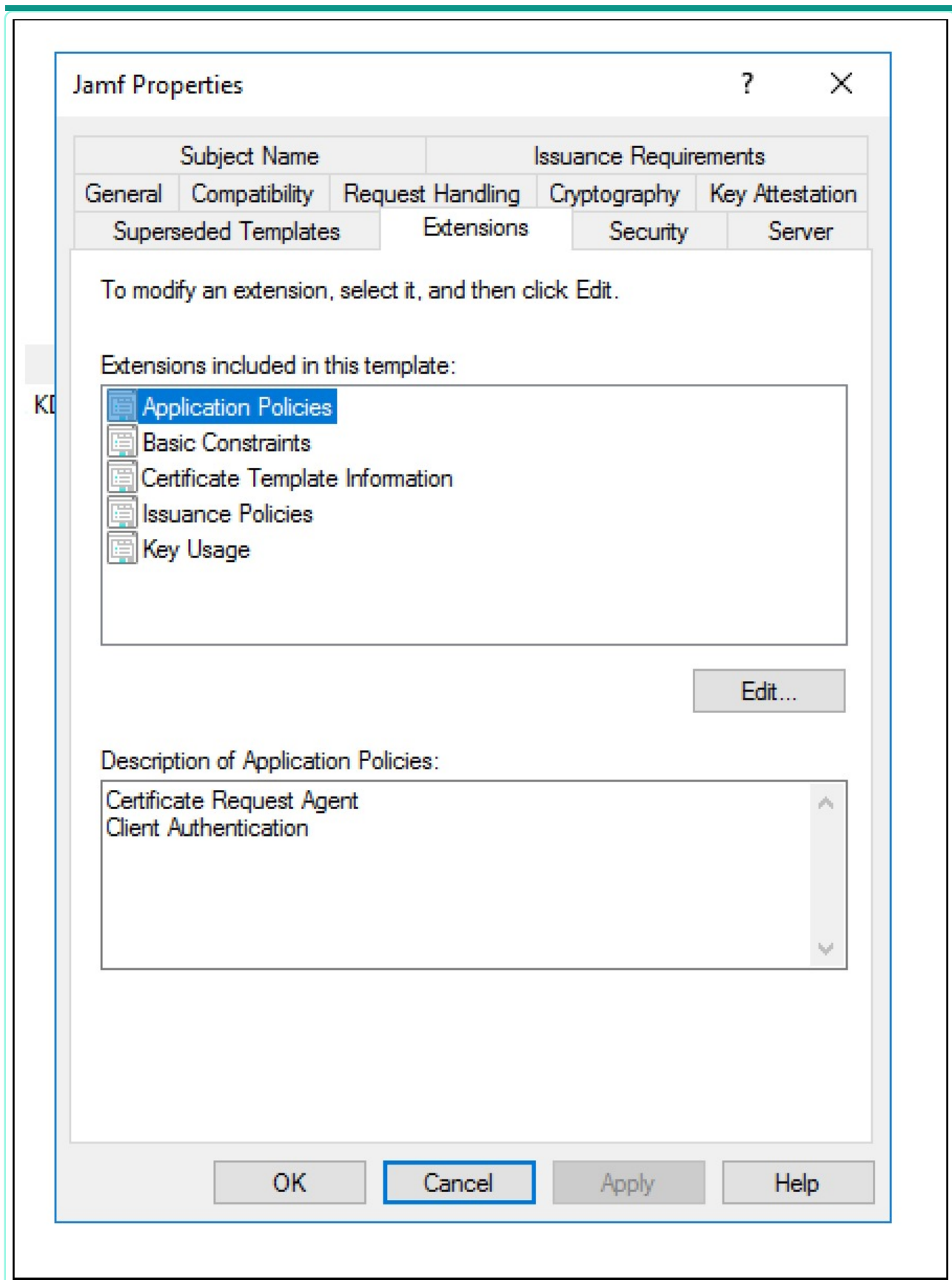
Key Attestation



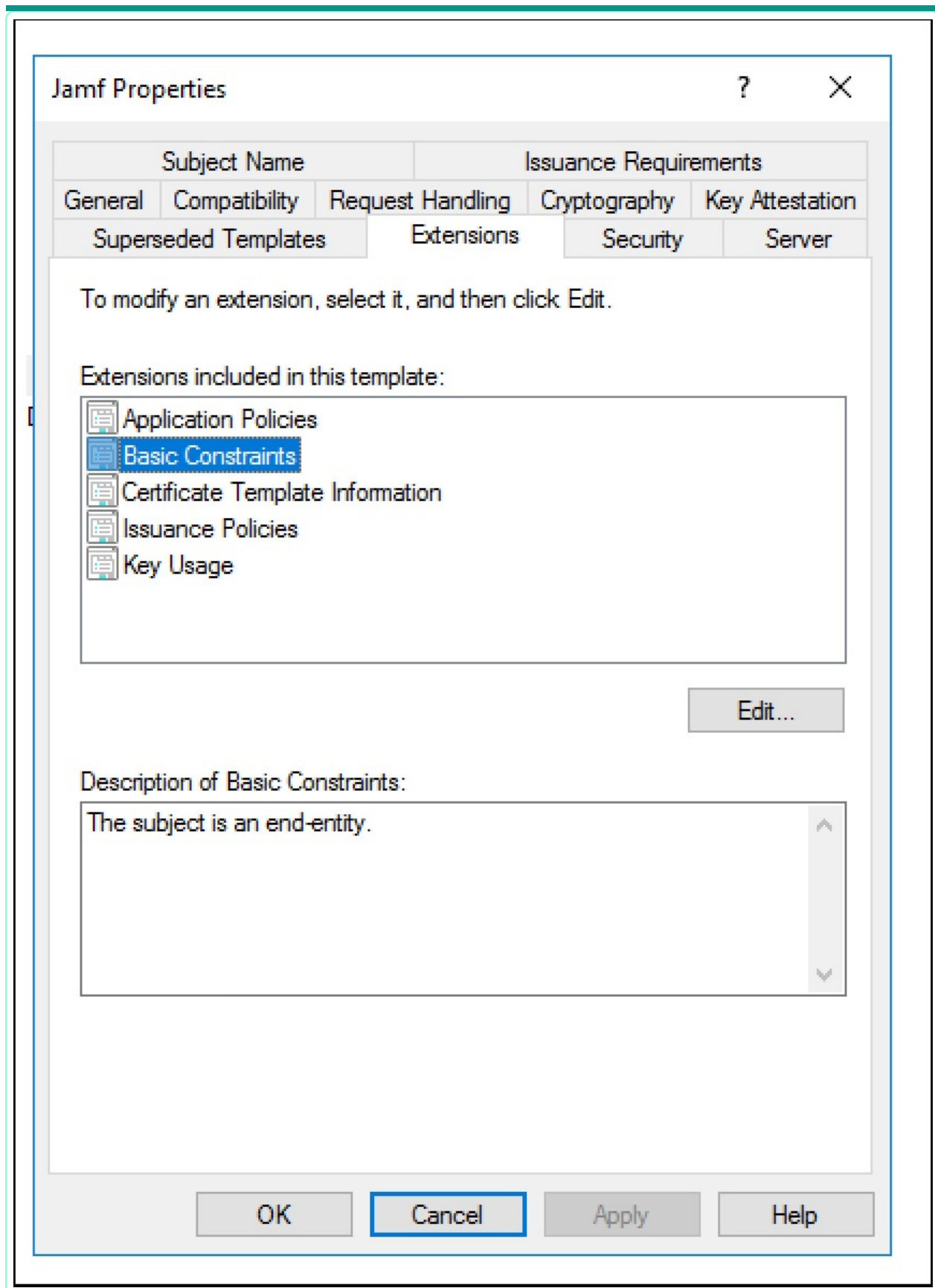
Superseded Templates



Extensions: Application Policies

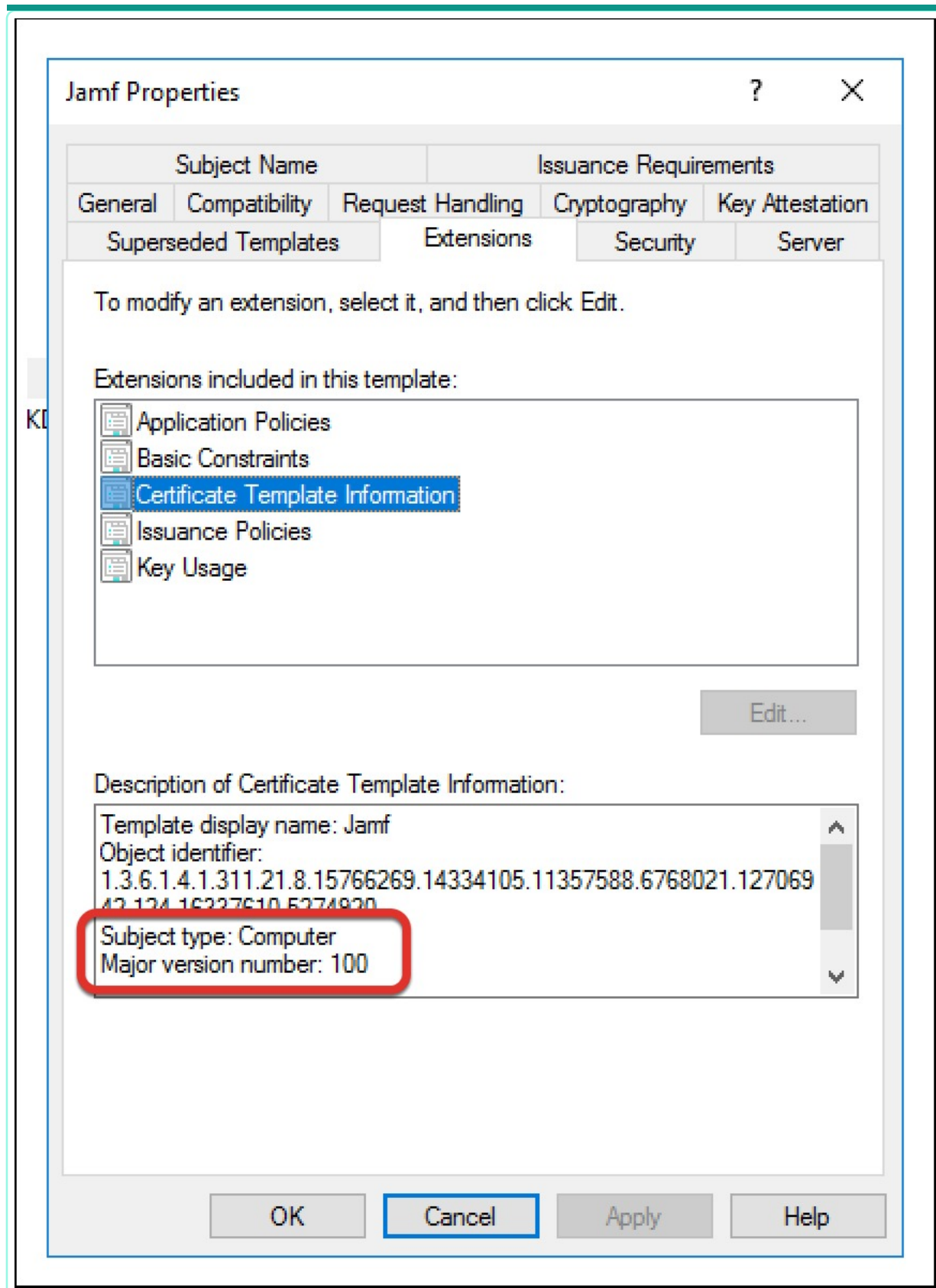


Extensions: Basic Constraints

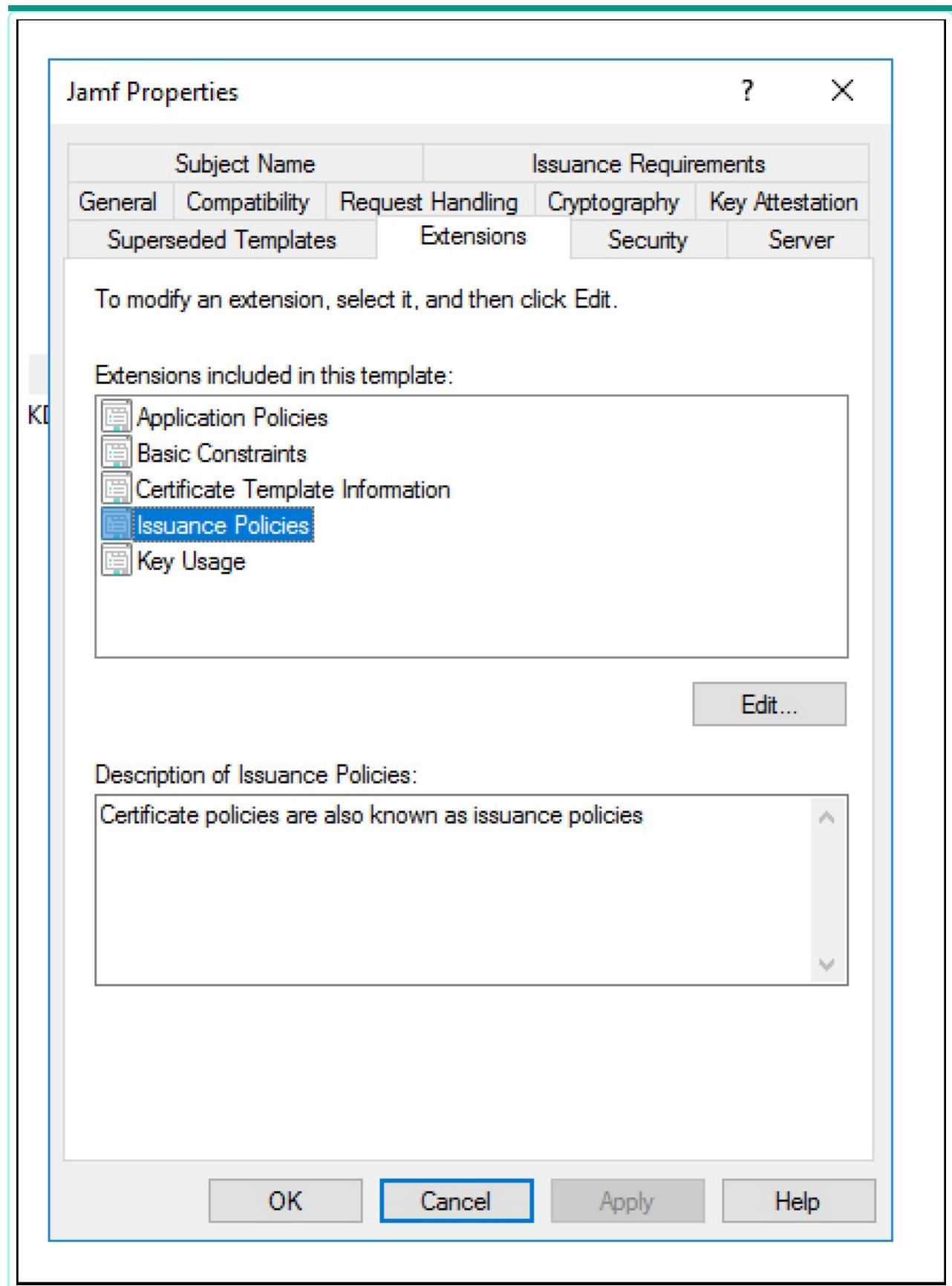


Extensions: Certificate Template Info

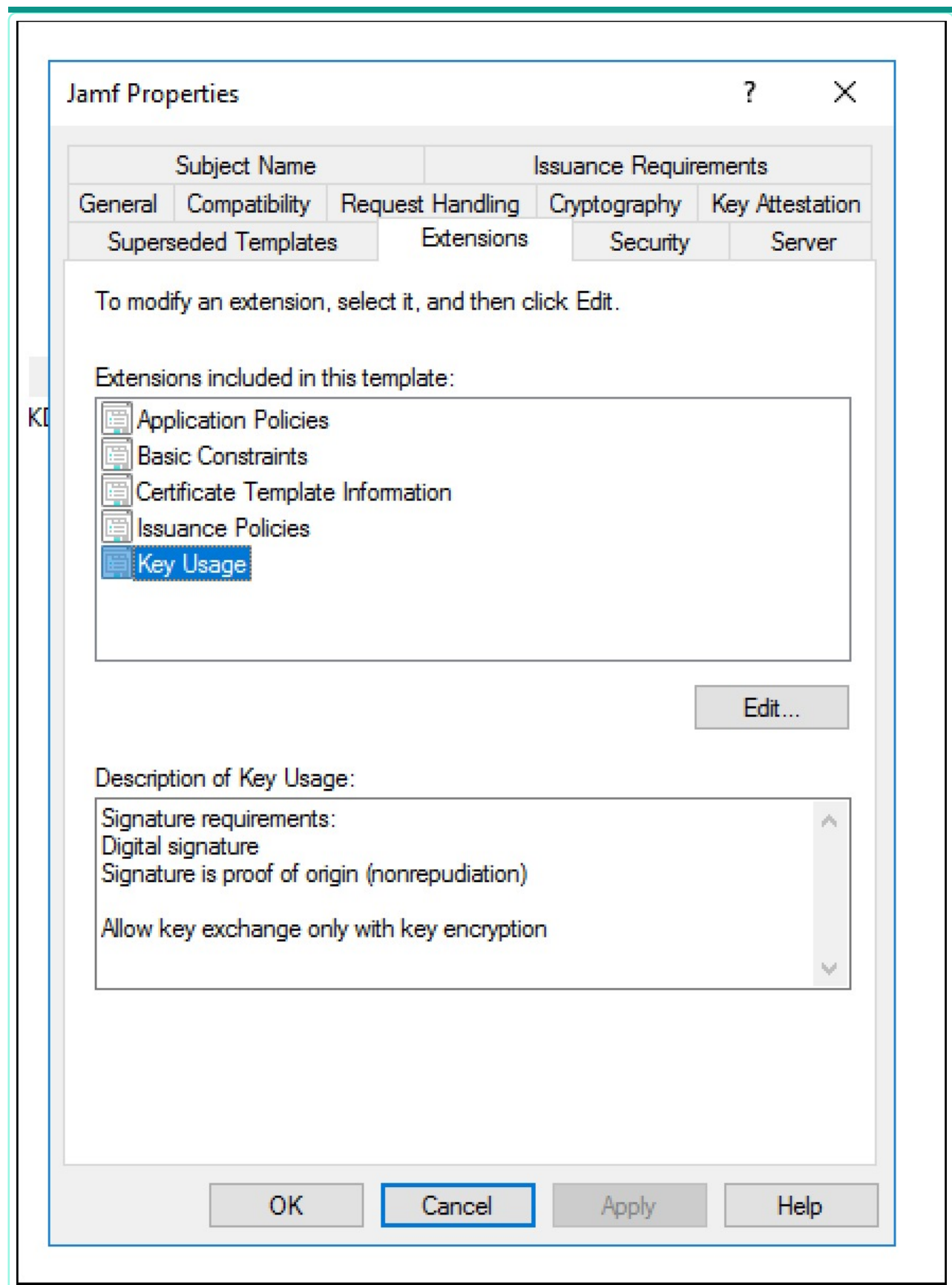
■ This is a computer template



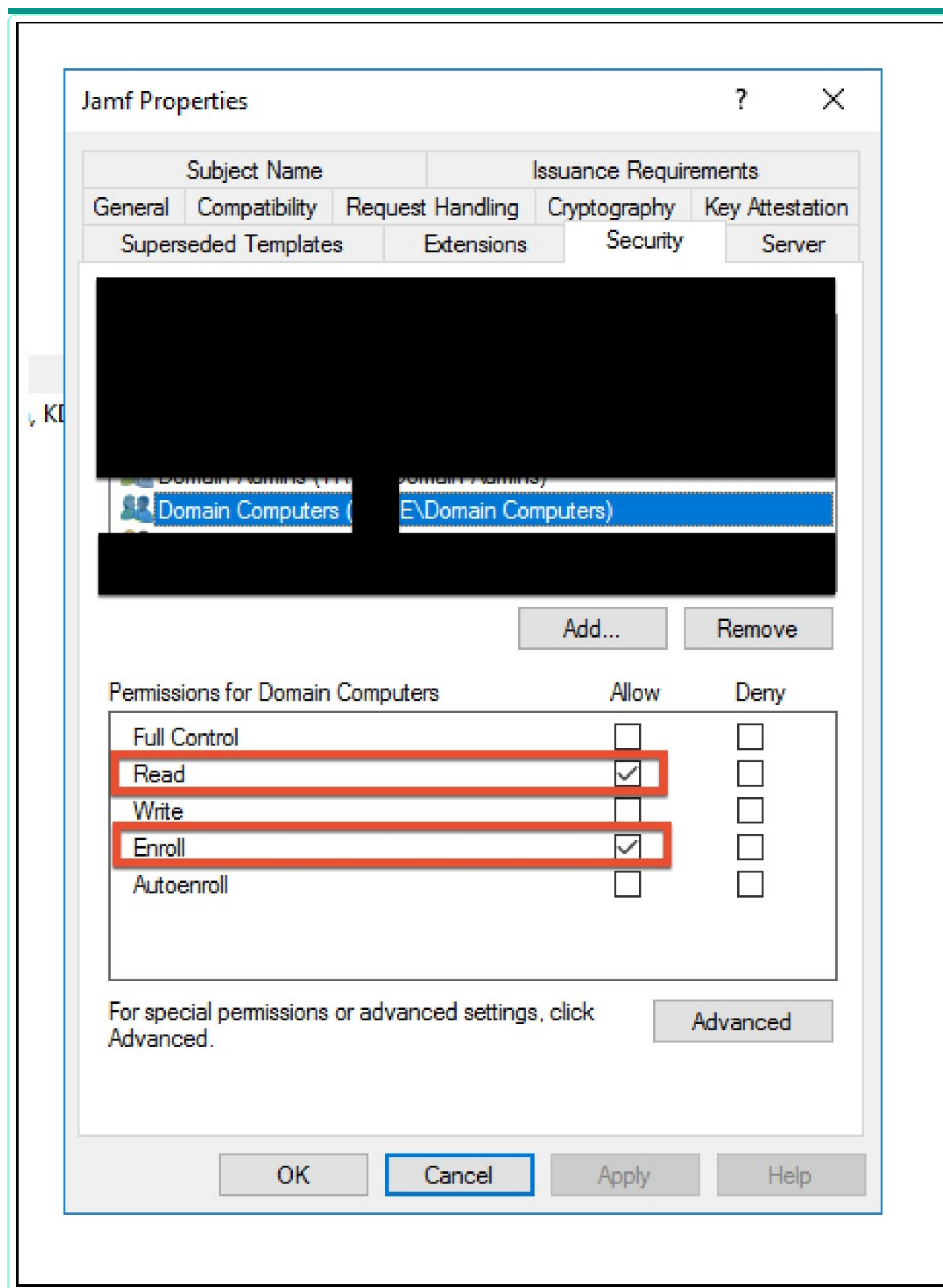
Extensions: Issuance Policies



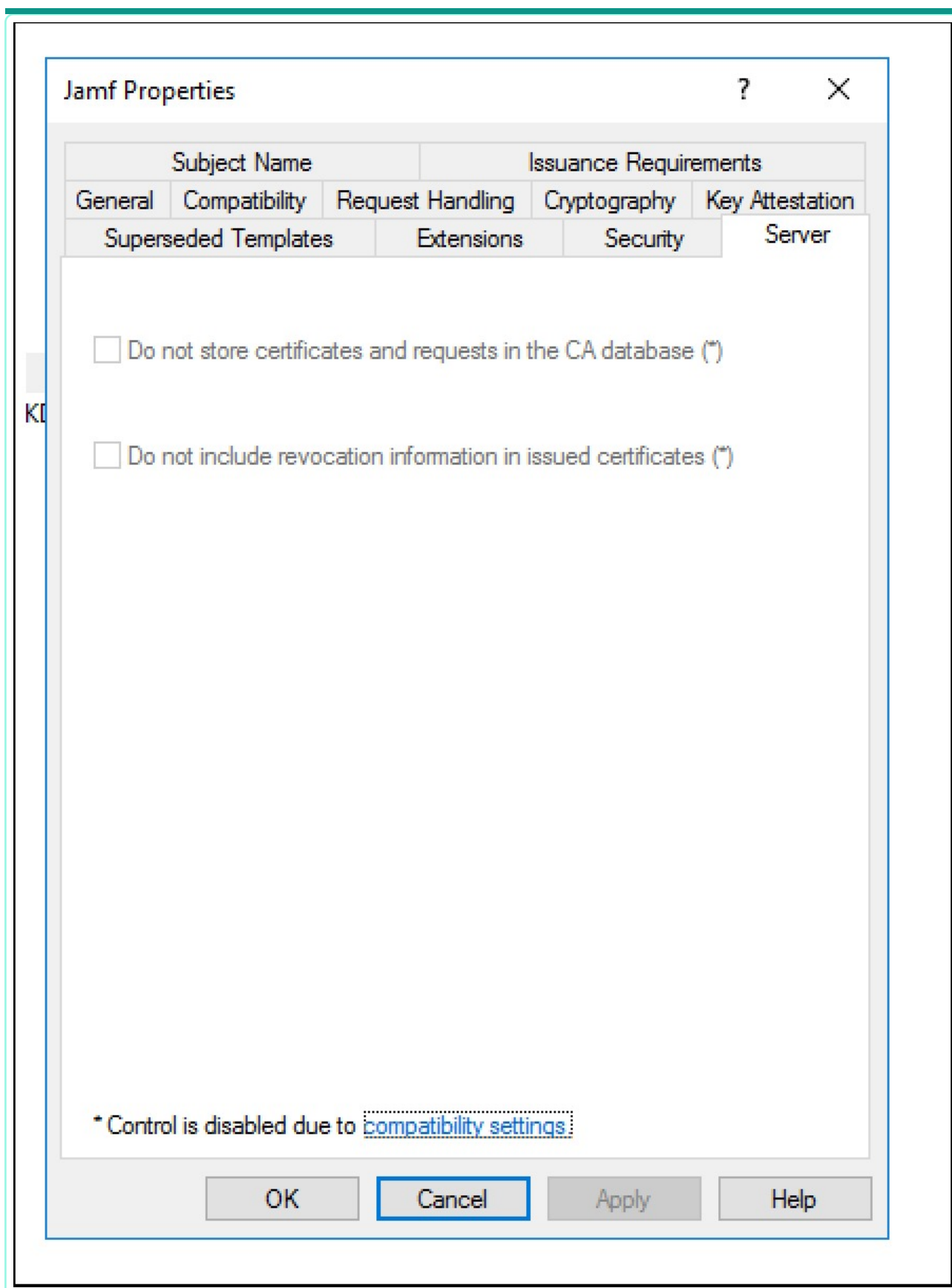
Extensions: Key Usage



Security



Server



Subject Name

Jamf Properties ? X

General Compatibility Request Handling Cryptography Key Attestation
Superseded Templates Extensions Security Server

Subject Name Issuance Requirements

☐ Supply in the request

☐ Use subject information from existing certificates for autoenrollment renewal requests (*)

☒ Build from this Active Directory information

Select this option to enforce consistency among subject names and to simplify certificate administration.

Subject name format:

Common name

☐ Include e-mail name in subject name

Include this information in alternate subject name:

☐ E-mail name

☒ DNS name

☒ User principal name (UPN)

☐ Service principal name (SPN)

* Control is disabled due to [compatibility settings](#).

OK Cancel Apply Help

Issuance Requirements

The screenshot shows the 'Jamf Properties' dialog box with the 'Issuance Requirements' tab selected. The dialog has a title bar with a question mark and a close button. Below the title bar is a tabbed interface with the following tabs: General, Compatibility, Request Handling, Cryptography, Key Attestation, Superseded Templates, Extensions, Security, and Server. The 'Issuance Requirements' tab is active, showing the following options:

- Require the following for enrollment:**
 - ☐ CA certificate manager approval
 - ☐ This number of authorized signatures:
- If you require more than one signature, autoenrollment is not allowed.
- Policy type required in signature:
- Application policy:
- Issuance policies:

Below the enrollment requirements, there are options for reenrollment:

- Require the following for reenrollment:**
 - ☒ Same criteria as for enrollment
 - ☐ Valid existing certificate
 - ☐ Allow key based renewal (*)
- Requires subject information to be provided within the certificate request.

At the bottom, a note states: * Control is disabled due to [compatibility settings](#).

The dialog has four buttons at the bottom: OK, Cancel (highlighted with a blue border), Apply, and Help.

Step 2: JAMF Setup

Corporate Wireless

Options

Scope

 General >

 Network
1 Payload Configured

 Certificate
2 Payloads Configured

 AD Certificate
1 Payload Configured

General

NAME Display name of the profile

Corporate Wireless

DESCRIPTION Brief explanation of the content or purpose of the profile

Creates cert to join Corporate

CATEGORY Category to add the profile to

Settings

DISTRIBUTION METHOD Method to use for distributing the profile

Install Automatically

LEVEL Level at which to apply the profile

Computer Level

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Network: Protocols

The screenshot displays the Jamf Pro Network configuration interface, specifically the 'Protocols' tab. The interface is divided into several sections:

- Options** and **Scope** tabs are at the top left.
- General** and **Network** (Physical Configured) are in the left sidebar.
- Network Interface** is set to 'Wi-Fi'.
- SERVICE SET IDENTIFIER (SSID)** is set to 'Corporate'.
- Hidden Network** is checked.
- Auto Join** is checked.
- PROXY SETUP** is set to 'None'.
- SECURITY TYPE** is set to 'WPA / WPA2 Enterprise'.
- Use as a Login Window configuration** is unchecked.
- Network Security Settings** are shown in a table with columns 'Protocols' and 'Trust'.
- SECURITY TYPE** is set to 'TLS'.
- Username** is set to 'Administrator'.
- TLS MINIMUM VERSION** is set to 'None'.
- TLS MAXIMUM VERSION** is set to 'None'.
- IDENTITY CERTIFICATE** is set to 'AD Certificate'.
- POST LANE QUALITY OF SERVICE (QoS) MARKING** is set to 'Mark All Apps'.

Network: Trust

The screenshot shows the 'Network: Trust' configuration page in Jamf Pro. The left sidebar contains 'Options' and 'Scope' tabs, with 'Options' selected. Below are links for 'General', 'Network', 'Certificate', 'AD Certificate', and 'AD Certificate'.

The main configuration area includes the following sections:

- NETWORK INTERFACE**: Type of network interface on the device. Set to 'WiFi'.
- SERVICE SET IDENTIFIER (SSID)**: Identification of the wireless network to connect to. Set to 'SSID'.
- Hidden Network**: ☐ (unchecked). Expects if target network is not open or broadcasting.
- Auto Join**: ☐ (unchecked). Automatically join this wireless network.
- PROXY SETUP**: Configures proxy settings to be used with this network. Set to 'None'.
- SECURITY TYPE**: Wireless network encryption to use when connecting. Set to 'WPA/WPA2 Enterprise'.
- Use as a Login Window configuration**: ☐ (unchecked). Allow sign-in for authentication to this network.
- Network Security Settings**: Configurations options for 802.1X network authentication. Includes a 'Protocols' dropdown set to 'Trust'.
- USERNAME**: Username for connection to the network.
- TLS MINIMUM VERSION**: Set to 'None'.
- TLS MAXIMUM VERSION**: Set to 'None'.
- IDENTITY CERTIFICATE**: Certificate for connection to the network. Set to 'AD Certificate'.
- TRUSTED CERTIFICATES**: Certificates trusted/expected for authentication. Includes 'Issuing 2' and 'Root CA'.
- PROTECTED SERVER CERTIFICATE NAMES**: Certificates for authentication to distribution server.
- CERTIFICATE COMMON NAME**: Includes 'Issuing 2' and 'Root CA'.
- Allow Trust Exceptions**: ☐ (unchecked). Allow a certificate to be trusted for use with this app.
- TRUST LANE QUALITY OF SERVICE (QoS) WARNINGS**: Set to 'Mark All Apps'.

Red boxes highlight the following elements:

- SECURITY TYPE: WPA/WPA2 Enterprise
- Protocols: Trust
- IDENTITY CERTIFICATE: AD Certificate
- TRUSTED CERTIFICATES: Issuing 2, Root CA
- CERTIFICATE COMMON NAME: Issuing 2, Root CA

Certificate

Computers > Configuration Profiles > [Redacted] Corporate Wireless

Options Scope

General

Network
1 Payload Configured

Certificate
2 Payloads Configured

AD Certificate
1 Payload Configured

Certificate

CERTIFICATE NAME Display name of the certificate credential
[Redacted] Root CA

CERTIFICATE X.509 certificate (.cer, .der, .p12, or .pfx) for inclusion on the device

FILENAME
[Redacted] Root.cer

ISSUER NAME
[Redacted] Root CA

EXPIRATION DATE
03/20/2057

PASSWORD Password used to secure certificate credentials

VERIFY PASSWORD

Certificate

CERTIFICATE NAME Display name of the certificate credential
[Redacted] Issuing 2

CERTIFICATE X.509 certificate (.cer, .der, .p12, or .pfx) for inclusion on the device

FILENAME
[Redacted] Issuing-2.cer

ISSUER NAME
[Redacted] Root CA

EXPIRATION DATE
[Redacted]

PASSWORD Password used to secure certificate credentials

VERIFY PASSWORD

AD Certificate

Options

Scope

General

Network
1 Payload Configured

Certificate
2 Payloads Configured

AD Certificate
1 Payload Configured

AD Certificate

DESCRIPTION

The description of the certificate request as shown in the certificate selector of other payloads such as VPN and Network

issuing CA 2

CERTIFICATE SERVER

The network address of the certificate server

c. .com

CERTIFICATE AUTHORITY

The name of the CA

issuing CA 2

CERTIFICATE TEMPLATE

The name of the certificate template, usually Machine or User

Jamf

CERTIFICATE EXPIRATION NOTIFICATION THRESHOLD

The number of days before the certificate expires at which to start showing the expiration notification

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☐ Prompt for credentials

Prompt the user for credentials. This setting is not supported for pushed profiles; the profile must be downloaded and manually distributed

USERNAME

The username with which to authenticate to the certificate server

PASSWORD

The password with which to authenticate to the certificate server

VERIFY PASSWORD

☐ Allow access to all applications

Allow all applications to access the certificate in the keychain

☐ Allow export from keychain

Allow admin to export private key from the keychain